

Conditional Query Answering over EHR

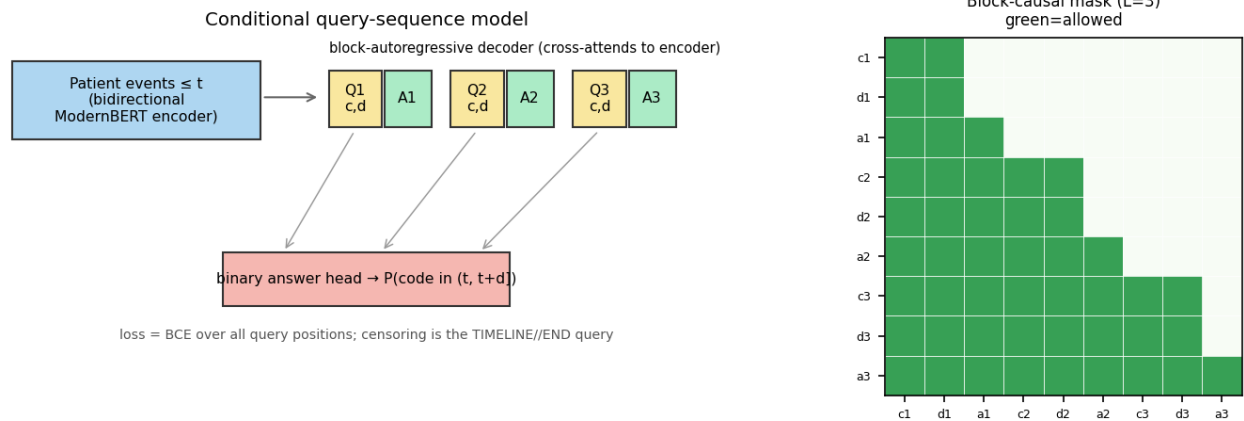
A block-autoregressive reformulation of EveryQuery, trained on MIMIC-IV (MEDS)

We rework EveryQuery (Chandak et al., arXiv:2603.07900) into a conditional query-sequence foundation model: a bidirectional encoder over the patient record feeds a block-autoregressive decoder that answers an ordered list of queries, each conditioned on the patient state and the teacher-forced answers of all earlier queries. Trained on a single NVIDIA GB10 over one no-repeat epoch of **28.79M** unique random query sequences (every step a fresh sequence). Held-out macro per-task AUROC is **0.792**; designed clinical tasks reach **0.859** AUROC (30-day mortality, post-admission), and conditioning on an intermediate ICU-admission query raises predicted 30-day mortality 2.8%→10.3%.

1. Task & model

Task. Each query is (code C , horizon d); the answer is binary — was C observed in $(t, t+d]$ for a patient at prediction time t ? The model answers an ordered sequence $[Q1][A1][Q2][A2]...$, predicting each A_j from the patient state, the queries $Q_{1..j}$, and the teacher-forced answers $A_{1..j-1}$ — never A_j itself. It thus learns $P(A_j | \text{patient}, Q_{1..j}, A_{1..j-1})$.

Architecture. A ModernBERT encoder (8 layers, hidden 384) embeds the tokenized MEDS history up to t . A 4-layer Transformer decoder runs over the query/answer stream and cross-attends to it under a block-causal mask: a query's code/duration tokens see each other and all tokens of earlier blocks (incl. their answers), but never the query's own answer and never later blocks. The answer for block j is read at its duration token. Encoder uses ModernBERT positions; the decoder adds learned block-position + token-type embeddings. ~33M parameters, bf16. Single binary head; loss is one BCE over all real query positions.



Censoring is a query, not a label. The end-of-timeline code TIMELINE//END (one per subject) is queried like any other: (END, d) = 'does the record end within d ?'. Conditioning a later query on its answer recovers and generalizes EveryQuery's implicit $P(\text{occurs} | \text{data after } d)$: the END=NO slice is the original question, while END=YES gives $P(\text{occurs} | \text{record ends})$ — the regime where terminal events like death actually fall.

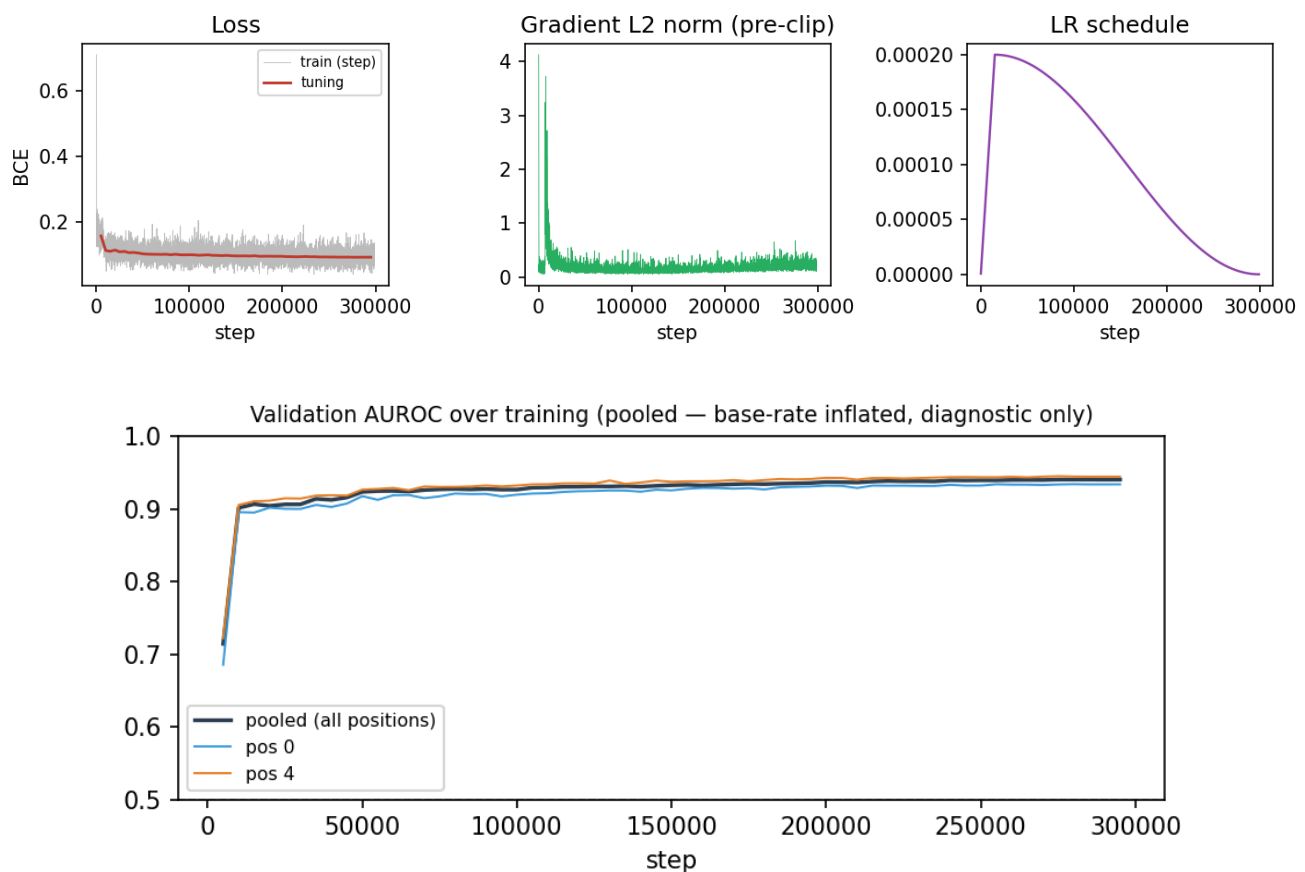
Leak post-mortem. An earlier design teacher-forced a same-horizon censor answer and masked censored labels; for terminal events the censor answer then equaled the label, and the model copied it — inflating 30-day mortality AUROC to 0.991 (0.996 from the censor answer alone). The binary occurrence + END-as-query design here removes that leak structurally; the honest mortality AUROC is 0.859 post-admission (§6).

2. Data & query sampling

MIMIC-IV in MEDS form (v0.3.0 tensorized cohort, 11,958 codes; 292 train / 37 tuning / 37 held-out shards). Training sequences are sampled fully at random: 5 i.i.d. queries per patient context (code uniform over the vocabulary incl. TIMELINE//END, horizon log-uniform over 1–365 days), with binary observed-occurrence labels from a single per-query asof join. We sampled **28.79M** unique sequences (86.9% distinct; all 227,602 subjects covered across all shards) and trained one no-repeat epoch — so every optimizer step sees a different query sequence (maximizing query-space coverage; verified no cross-shard query replication). Fixed length 5 is used because, under the block-causal mask, position j depends only on blocks $\leq j$, so shorter sequences are redundant prefixes — fixed length trains every position with uniform sample counts.

3. Training stability & convergence

AdamW (lr $2e-4$, cosine schedule, 5% warmup), grad-clip 1.0, bf16, batch 96, 298,999 steps (~22.5 h). Training was stable throughout: the pre-clip gradient norm stayed bounded (median 0.15, max 4.13) with no loss spikes; tuning loss fell smoothly to 0.092 and plateaued, indicating clean convergence on this one-epoch schedule.



4. Evaluation methodology

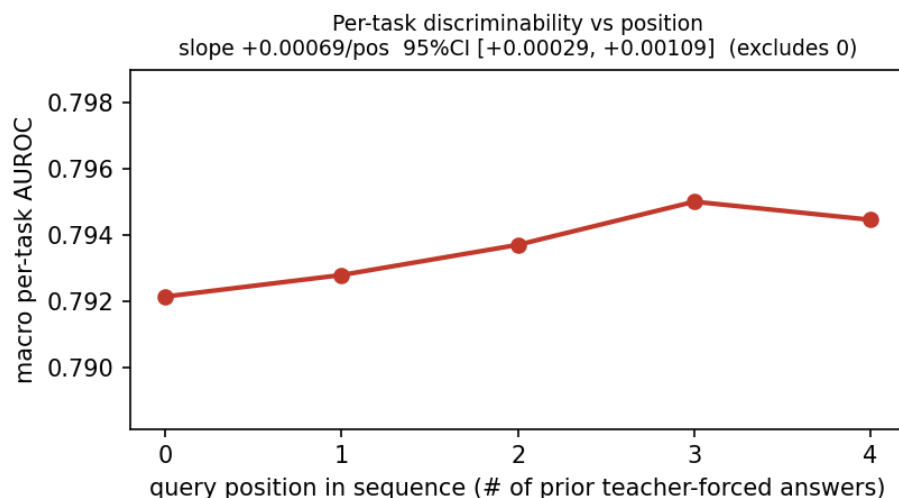
Macro per-task AUROC, not pooled. Pooled AUROC scores cross-task pairs (a positive for one query vs a negative for another) and is dominated by base-rate differences between queries — it overstates per-query skill (pooled ≈ 0.95 here). The right metric is AUROC computed *within each query*, macro-averaged. Since $\text{AUROC} = P(\text{score}_{\text{pos}} > \text{score}_{\text{neg}})$ (Mann–Whitney), for one positive/negative patient pair drawn for the same query the indicator $1[\text{score}_{\text{pos}} > \text{score}_{\text{neg}}]$ is an unbiased estimate of that query's AUROC, and averaging over many queries estimates macro-AUROC.

Positives are constructed **occurrence-driven** — take a real occurrence of C at τ and a prediction time $t \in [\tau-d, \tau)$ — so the label is guaranteed positive and *all codes are estimable, rare included* (no coverage bias toward common codes). We use the **patient-uniform** scheme (each patient that permits a positive picked uniformly, then one of its positive times) so long stays don't dominate. For the position trend, the *same* positive/negative pair is scored at every position (fillers' true answers teacher-forced), so the comparison is fully paired and any sampling bias cancels; we bootstrap over tasks for the slope CI.

5. Results — per-task discrimination & the conditioning trend

position (priors)	macro per-task AUROC
0	0.7921
1	0.7928
2	0.7937
3	0.7950
4	0.7945

On 38,330 held-out tasks (patient-uniform), per-task AUROC is **0.792** at position 0 and rises monotonically with position. The trend is small but statistically significant: slope **0.00069/position**, 95% CI [0.00029, 0.00109] (excludes 0); Spearman $\rho = 0.90$ [0.50, 1.00]. Later queries, which condition on more prior (query, answer) context, are answered measurably better — confirming the model uses the conditional structure (lower Bayes error with more information). The increment is tiny (+2.3 milli-AUROC pos 0→4): random priors are only weakly informative about a random target, and a well-trained encoder already captures much of the signal, leaving little for the priors to add.



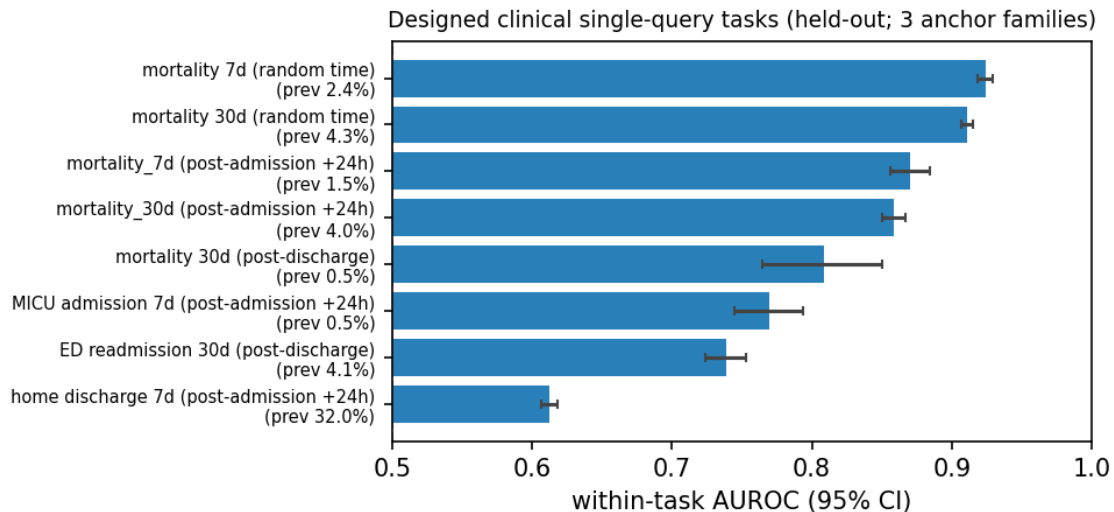
5b. Comparison to original EveryQuery (uncensored cohort)

Original EveryQuery reports occurrence AUC only where censoring is false (the window is fully observed). We replicate that exactly — task sequence `[TIMELINE//END, d]=0 [C, d]` evaluated only on contexts where the record does not end within `d` — over 36,890 tasks. Macro occurs-AUROC is **0.794** [0.790, 0.798] with the EOS=NO prefix, vs 0.793 without it — statistically indistinguishable (the cohort is already uncensored, so the prefix adds little). This ~0.79 is the apples-to-apples number against original EveryQuery's headline metric; note terminal codes (death) have no uncensored positives by construction and are absent here — exactly original EveryQuery's blind spot, which our model covers via the EOS=YES (record-ends) regime.

6. Results — designed clinical tasks

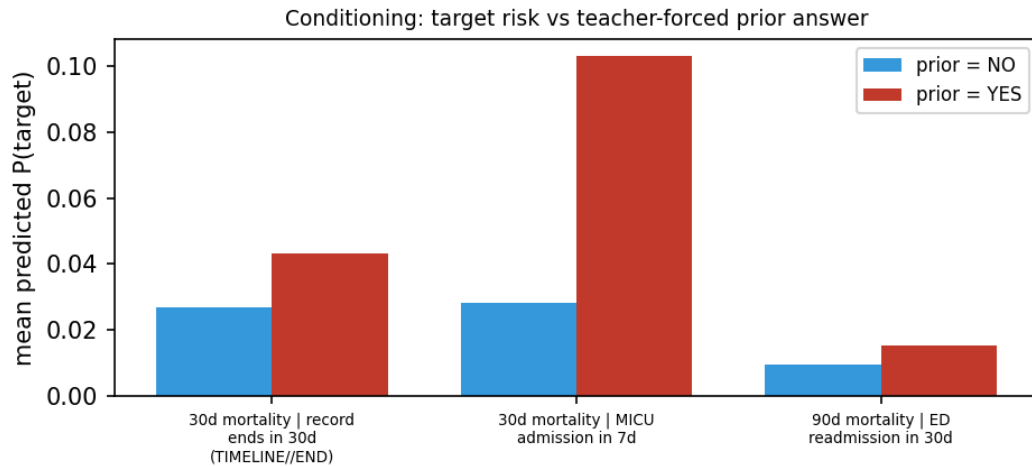
We designed clinically meaningful single-code tasks over three held-out anchor families — post-admission (24h after a HOSPITAL_ADMISSION; 37,907 anchors), post-discharge (at a HOSPITAL_DISCHARGE//HOME event; 16,563), and random-time (a uniformly sampled valid event time; 86,127) — each requiring ≥ 10 prior events and ≤ 3 anchors/subject. Readmission is anchored at discharge (a genuine post-discharge metric); each task names its exact single MEDS code. Within-task AUROC, bootstrap 95% CI.

clinical task	prevalence	AUROC	95% CI
mortality 7d (random time)	2.42%	0.924	[0.918, 0.929]
mortality 30d (random time)	4.30%	0.911	[0.907, 0.915]
mortality_7d (post-admission +24h)	1.54%	0.870	[0.856, 0.884]
mortality_30d (post-admission +24h)	4.03%	0.859	[0.850, 0.867]
mortality 30d (post-discharge)	0.50%	0.809	[0.765, 0.850]
MICU admission 7d (post-admission +24h)	0.50%	0.770	[0.744, 0.794]
ED readmission 30d (post-discharge)	4.08%	0.738	[0.724, 0.753]
home discharge 7d (post-admission +24h)	32.04%	0.612	[0.606, 0.618]



Conditioning demonstrations. Scoring a target query under a teacher-forced prior answer (forced YES vs NO on the same anchors) shows the model updates downstream risk in a clinically sensible direction — the capability the block-autoregressive form adds:

conditional query (target prior)	P prior=NO	P prior=YES	natural AUROC
30d mortality record ends in 30d (TIMELINE//END)	0.027	0.043	0.904
30d mortality MICU admission in 7d	0.028	0.103	0.861
90d mortality ED readmission in 30d	0.010	0.015	0.835



The clearest example: teacher-forcing 'MICU admission within 7 days' raises predicted 30-day mortality from 2.8% to 10.3% ($\approx 3.6\times$) — the model correctly treats an impending ICU transfer as a strong mortality signal.

7. Discussion

A single model answers arbitrary queries well. Trained only on random query sequences (no task-specific supervision), the model reaches honest macro within-task AUROC ~ 0.79 across tens of thousands of held-out (code, horizon) tasks (and the same ~ 0.79 on original-EveryQuery's uncensored-occurs cohort), and 0.81–0.92 on designed mortality tasks across anchor families — leak-free, unlike the 0.99 of the flawed earlier design.

Conditioning works, and is used when it carries signal. Per-task AUROC rises monotonically and significantly with sequence position (more priors $\rightarrow$ lower Bayes error), and clinically the model sharply updates downstream risk on informative priors (ICU $\rightarrow$ death $3.6\times$). The position effect is small in aggregate because random priors are weakly informative and a strong encoder is partly redundant with them — but it is real and statistically significant.

Censoring without a censor head. Expressing censoring as the TIMELINE//END query makes the model strictly more expressive than EveryQuery (it can answer $P(\text{occurs} \mid \text{record ends})$, not just $P(\text{occurs} \mid \text{data continue})$) and removes the terminal-event leak by construction.

Limitations. Evaluation is teacher-forced (conditional calibration, not free-running rollout). The macro metric covers codes with an estimable positive rate. Sequences are length 5; longer-context behavior and free-running multi-step querying are natural next steps.

Code & docs: github.com/mmcdermott/EveryQuery-conditional (main). CLIs: `EQ_generate_query_sequences`, `EQ_predict_sequences`, `EQ_evaluate_sequences`. Evaluation: `scripts/eval_macro_position.py` (per-task), `scripts/eval_clinical.py` (clinical). See `CONDITIONAL_QUERIES.md`.